

Turbulence modulation via dilute surfactant additives in experimental Taylor–Green flow

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We present an experimental investigation of the turbulence modulation caused by dilute solutions of cationic surfactant (CTAC/NaSal) in a Taylor–Green-like flow. By driving the flow with rectangular blades, we isolate the inertial forcing and analyze the effect of surfactant addition on both the mean flow and velocity fluctuations across a wide range of Weissenberg and Reynolds numbers. In contrast to the classical picture of monotonic turbulence attenuation past a critical Weissenberg number, we show that surfactant addition produces distinct regimes as the rotation rate is varied. At low rotation rates, the mean flow and velocity fluctuations are strongly intensified through a resonance mechanism between the periodic forcing imposed by the blades and the formation of large-scale micellar structures. Spatially resolved measurements in the region between the blades reveal that this mechanism is mediated by the formation of jet-like structures which resemble the shear-induced structures reported in wall-bounded drag reduction studies. Once this periodic contribution is removed via triple decomposition of the velocity field, the resulting turbulent fluctuations recover part of the classical picture, with a characteristic attenuation of turbulent energy once the Weissenberg number, estimated from the Kolmogorov time scale of the Newtonian flow, surpasses the value of unity. As the rotation rate is further increased, the turbulence attenuation progressively vanishes as the micellar network breaks down under the strong shear. The obtained results indicate that the modulation of bulk turbulence under dilute surfactant solutions cannot be captured by a single relaxation time alone, and instead involves at least two groups of viscoelastic structures with distinct length and time scales, whose interplay is strongly sensitive to the nature of the turbulent forcing.

I. INTRODUCTION

Since the discovery that trace amounts of dissolved polymer could lead to drag reduction in pipe flows [1], the use of additives for drag reduction in wall-bounded flows has become well established. Besides polymer solutions, it has since been discovered that self-assembling surfactant solutions share similar effects [2], owing to the fact that dilute surfactant solutions can form wormlike micelles whose relaxation dynamics mimic that of polymer chains while resisting mechanical degradation.

Beyond the applications of drag reduction, it has been shown that viscoelastic additives modulate turbulence more broadly, reshaping the energy spectrum and producing an elastic range governed by additive elasticity rather than inertia at the small scales [3, 4], with surfactant-specific micellar structures shown to alter vortex shedding and turbulence statistics directly [5]. The classical picture underlying most of these observations is a time-scale matching argument: turbulent eddies with a turnover time shorter than the relaxation time of the additive interact appreciably with it, leading to a truncation of the energy cascade below a corresponding length scale, the Lumley scale [2, 6], with a refined version of this argument predicting a somewhat smaller truncation scale on the basis of an elastic energy budget between the additive and the turbulent eddies [7]. This time-scale matching is conveniently expressed through the Weissenberg number,

$$Wi \equiv \frac{\tau_s}{\tau_n}, \quad (1)$$

defined as the ratio between the relaxation time τ_s of

the viscoelastic solution and a characteristic timescale τ_n of the turbulent flow, most commonly taken as the Kolmogorov timescale τ_η of the Newtonian flow. Broadly speaking, turbulence modulation by the additive is expected to become significant once $Wi \gtrsim 1$, since only then do turbulent eddies become slow enough for the additive to respond elastically rather than viscously.

In spite of academic and commercial interest, comparatively few studies address the behavior of dilute surfactant solutions in the context of three-dimensional bulk turbulence, with most of the literature focused on wall-bounded or quasi-two-dimensional flows. A notable exception, and one directly relevant to the present work, is the study of Horimoto and Goto [8], who investigated the modulation of confined turbulence sustained by the precession of a spherical container using dilute solutions of both CTAC and polyethylene oxide (PEO). Using the same Lumley-scale/elastic-cascade arguments described above, they showed that small-scale turbulent eddies are drastically suppressed by the CTAC additive while a pair of large-scale coherent vortex tubes, responsible for sustaining the turbulence through an energy cascade, survives essentially unmodified. This result illustrates both the strength of the classical time-scale matching argument in predicting which additive is effective in a given flow, and the fact that the interaction between additive elasticity and large-scale, coherent flow structures (as opposed to the small-scale turbulent cascade alone) can play a central role in determining the overall response of the flow. However, since the precessing sphere used in that study generates a comparatively steady, nearly axisymmetric large-scale flow, it remains an open question how

this picture extends to a flow driven by an explicitly periodic, spatially-localized forcing, where the additive can interact with both the mean flow and the periodic forcing itself, in addition to the turbulent cascade.

This gap is compounded by the fact that the rheology of very dilute surfactant solutions remains difficult to pin down, with reported values of viscosity and relaxation time of CTAC/NaSal solutions (such as the one considered in the present experiment) varying considerably between studies [9, 10]. In spite of the diverging measurements and reports that some of the dilute solutions behave as a Newtonian fluid under rheological measurements [9], such solutions are known to produce measurable drag reduction [11], which further reinforces the need to analyze such solutions more in depth.

Motivated by this gap, here we present an experimental analysis of the turbulence generated by a four-rotor, Taylor–Green-like experiment using dilute CTAC solutions, in an attempt to characterize the modulation of turbulence caused by the addition of the surfactant across a wide range of Reynolds numbers, and consequently Weissenberg numbers, for flows driven by inertial forcing. In contrast to the precessing-sphere geometry discussed above, the current apparatus subjects the flow to an explicit periodic forcing at the scale of the rotor spacing, allowing us to probe not only the classical attenuation of the turbulent cascade, but also the interaction of the surfactant’s elasticity with the mean flow and the periodicity of the forcing itself.

II. EXPERIMENTAL DEVICE

In the present study, we employ an experimental device designed to produce Taylor–Green-like forced turbulence in a laboratory setting. The apparatus, with schematic shown in figure 1, consists of a water tank with four rotors which are driven by a gear system and rotate in opposing directions in order to produce turbulence in the region between them (hereinafter simply referred to as the bulk), which is the target of our studies. Spatially-resolved measurements are obtained via 2D particle tracking (briefly described in section II A) using a traditional laser-sheet and high-speed camera combination, which are positioned as also shown in figure 1 for the case of bulk imaging.

Aside from the imaging of the bulk region, it is also possible to change the relative orientations of the laser sheet and camera, and use a mirror to allow for top-down imaging of the region in the vicinity of the blades, as shown in the schematic of figure 2.

As shown in the schematics of figures 1 and 2, the rotors themselves are shaped as blades with rectangular cross-section. While the design of the experimental device allows for the geometry of the rotors to be changed between cylinders and the carved blades, the use of the latter allows us to isolate inertial forcing in the driving of the flow, and therefore ignore boundary-layer effects

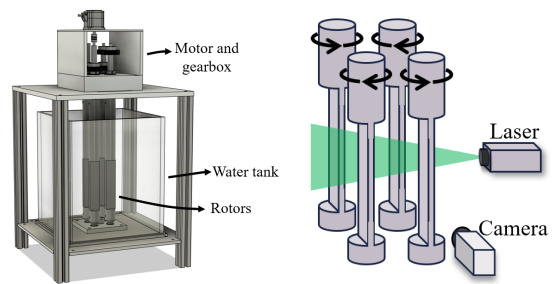


FIG. 1. CAD of the experimental device (left) and schematic showing normal orientation of camera and laser sheet, as well as direction of rotation of the rotors.

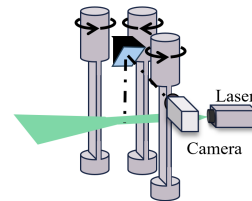


FIG. 2. Schematic of the altered orientation of camera and laser using mirror for imaging of the space between rotors. One of the rotors is omitted for clarity.

which would be present in the case of viscous driving which happens when cylindrical rotors are used. Additionally, the rotation direction is such that mean stretching of the flow happens in the out-of-plane component of the camera image (when imaging the bulk region in the vertical plane), and is accompanied by the formation of secondary vortices as shown in previous numerical studies [12]. For simplicity, the carved rotors will hereinafter simply be referred to as blades.

While not shown in the schematics, the experimental device is also encased in thermal insulation and connected to an air-processor system. By controlling the temperature of the surrounding air, we maintain the water temperature at approximately 26 °C over the course of the experiments.

A. Data measurement and reduction

As previously mentioned, processing of the obtained images to calculate the velocity fields is conducted via a particle tracking velocimetry (PTV) code developed in-house. Although the details of the algorithm are out of the scope of the current document, the code is built as a collection of previously established methods. For the detection of tracer particles from the images, we employ a variation of the Iterative Particle Reconstruction (IPR) method, which is commonly employed for volumetric particle tracking as part of the Shake-the-Box method [13]. The tracking of detected particle positions between the video frames is carried out by a variation of the Polar

Coordinate-System Similarity (PCSS) method [14].

After tracks have been identified, trajectories are filtered first with a common motion estimator to remove camera vibrations, and then according to a Wiener Filter [15], which is calculated from the position spectra of the particles in a collection of trajectories. This then allows us to extract the noise floor of position estimation and determine the cutoff frequency of the filter for each of the cases. After the position of particles has been filtered, velocities are then obtained via second-order finite differencing. While two-point statistics such as the velocity structure functions are obtained directly from the Lagrangian data through a spatial binning approach, estimation of mean flow and velocity fluctuation RMS across the measurement domain is conducted after velocities have been interpolated into a regular grid through adaptive Gaussian interpolation [16].

The confidence intervals of the results for mean flow and velocity fluctuations, when presented, are estimated using the Seasonal Block Bootstrap (SBB) method [17], which consists of a bootstrap method where the data blocks are aligned with the forcing period (in this case the rotation of the rotors) to preserve the phase-locked structure of the results. A block length of 20 cycles, or 10 full rotations of the rotors, was selected through a sensitivity sweep study, and confidence intervals are computed using the percentile method.

B. Surfactant solutions

The surfactant used in the current study was the cationic surfactant cetrimonium chloride (CTAC), with the addition of sodium salicylate at a molar ratio of approximately 4:1 to provide counterions. The solution is prepared by mixing the adequate amounts of surfactant and salt with tap water and then agitating in a magnetic stirrer over a period of at least 12 hours, after which the resulting solution is added to the experimental device. For simplicity, the surfactant and salt solution is hereinafter referred to simply as the CTAC solution, with the solution concentration corresponding exclusively to the concentration of the surfactant counterpart.

For the analysis of the effect of solution concentration on the results, three CTAC solutions of concentrations of 33 ppm, 50 ppm, and 100 ppm were prepared. As previously mentioned, rheological measurements of such dilute solutions have historically yielded diverging results [9, 10], meaning that, to the best of the authors' knowledge, there is no precise measurement of properties such as the relaxation time of the micellar structures under these conditions.

Moreover, the molar ratio of 4:1 chosen for the current study differs from that of the aforementioned measurements, as well as that of drag reduction studies [11]. Although numerical studies have reported that such high molar ratios can lead to the formation of branched structures and a loss of viscoelasticity [18], such claims have

been made for solutions of much higher concentrations. In the authors' experience, this effect is not as pronounced in the case of dilute solutions, and the use of a high molar ratio was chosen to favor the formation of longer micellar structures.

In spite of these divergences, it can be expected from the aforementioned studies that the relaxation time of the solution, τ_s , should be of the order of 10^{-1} seconds. Although τ_s is expected to vary with the concentration of the solution, the same studies indicate that this variation is not necessarily monotonic, and is in any case small enough that τ_s should remain of the same order of magnitude across the three concentrations considered here. We therefore adopt the representative value $\tau_s = 0.1$ s as the point at which $Wi \approx 1$ for all three solutions.

Since τ_s cannot be precisely known beforehand, or independently controlled over a wide range, a sweep over the Weissenberg number Wi is instead achieved by changing the Reynolds number of the generated turbulence. For each of the prepared solutions, the rotation rate of the blades is changed from 2^{-3} to 2^4 Hz in powers of two; for the 100 ppm solution specifically, two additional, lower rotation rates of 2^{-4} and 2^{-5} Hz were also tested, in order to further resolve the behavior of the flow in the regime of very low Wi . Although it is debatable whether the flow generated at the lower rotation rates can be termed turbulent, it remains chaotic and unsteady, so such cases are still useful for determining the effect of surfactant addition on this type of unsteady, low-Reynolds-number flow. With this consideration, the range of Weissenberg numbers obtainable in the current experiment spans from $Wi = \mathcal{O}(10^{-2})$ to $Wi = \mathcal{O}(10^2)$, allowing us to observe the transition to the attenuated regime regardless of small changes in the precise value of the relaxation time.

The next section briefly describes the characterization of the turbulence generated in the current experimental device for the case of a Newtonian fluid.

III. TURBULENCE CHARACTERIZATION

Considering the unique nature of the current experimental apparatus, we conduct a characterization of the turbulence generated in the case of Newtonian flow as a function of the rotation rate of the blades. Although the generated turbulence is anisotropic, turbulent statistics become approximately homogeneous in the vicinity of the stagnation point of the mean flow in the center of the experiment. Owing to this, the statistics are computed as spatiotemporal averages over the duration of the experiments. Figure 3 displays the velocity fluctuation RMS as a function of the blade rotation speed, demonstrating that there is an approximately linear trend, with the ratio between the two components remaining somewhat constant as well. This trend is consistent with a simple dimensional argument: since the geometry of the apparatus is fixed, the only velocity scale available to the

large, forced eddies is the blade tip speed, $u' \sim \Omega R$, with R a characteristic length of the rotors. Note that the turbulent intensity is more pronounced in the horizontal direction, consistent with the primary forcing direction of the blades.

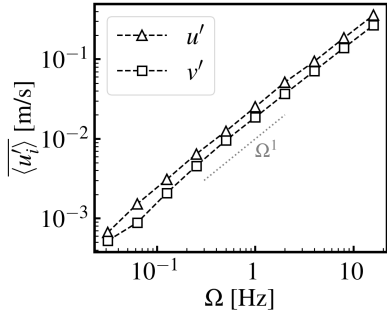


FIG. 3. Spatially averaged velocity fluctuation RMS in the bulk as a function of rotation rate of the agitator blades.

Although direct measurement of the dissipation range is not feasible under current experimental conditions, the turbulent dissipation can be estimated through the third-order structure function, as shown in figure 4. Although the large scales of the generated turbulence are anisotropic, the results of figure 4 indicate that the dissipation rate is approximately independent of the measurement direction. This is consistent with Kolmogorov's local isotropy hypothesis, according to which the small, dissipative scales of a turbulent flow are expected to recover statistical isotropy even when the large, energy-containing scales are not, and lends support to the validity of the obtained estimate. In all the plots, the increase in the value of the structure function as the separation distance reaches the smallest values is caused by the finite depth-of-correlation and the nature of planar imaging, as well as the fact that no interpolation of the instantaneous velocity fields is used for the calculation of these statistics.

For the cases of high rotation rate, the appearance of a horizontal region indicates that an inertial range is well-defined, and the dissipation rate can be taken as the average across this region. For the lower rotation rates, the Reynolds number is too low for the appearance of a defined inertial range, and the extraction of a representative value of dissipation is more difficult. For consistency between the cases, we consider the average value of the normalized third-order structure function between the separations of $r = 3$ mm and $r = 20$ mm as the dissipation value. These representative values are represented as the dotted lines of figure 4.

Combined with the previous scaling argument for u' , this measurement of the dissipation rate allows us to estimate its dependence on rotation rate. In fully developed turbulence, the dissipation rate becomes independent of viscosity and is instead set by the rate at which energy is supplied to the large scales, $\varepsilon \sim u'^3/L$, with L a fixed characteristic length of the energy-containing eddies

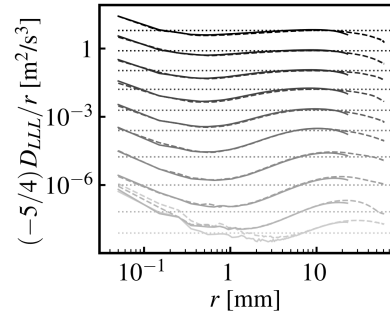


FIG. 4. Normalized third-order structure function for different rotation rates. Darker lines represent a higher rotation rate.

(again set by the geometry of the apparatus). Combined with $u' \sim \Omega R$, this yields the scaling $\varepsilon \sim \Omega^3$. From these results, we can then obtain the approximate Kolmogorov length scale and time scale for each of the cases,

$$\eta = \left(\frac{\nu^3}{\varepsilon} \right)^{1/4} \sim \Omega^{-3/4}, \quad \tau_\eta = \left(\frac{\nu}{\varepsilon} \right)^{1/2} \sim \Omega^{-3/2}, \quad (2)$$

as shown in figure 5, where the measured power-law trends are consistent with the scaling above. The estimated Kolmogorov length scale provides us with an additional clue as to why it is not possible to directly measure the dissipation range, considering that these scales become smaller than the depth-of-correlation of the current imaging system for most rotation rates. Additionally, the wide range spanned by the estimated time scales indicates that a correspondingly wide range of Weissenberg numbers should be obtained regardless of the fact that the relaxation time of the surfactant solutions is not precisely known.

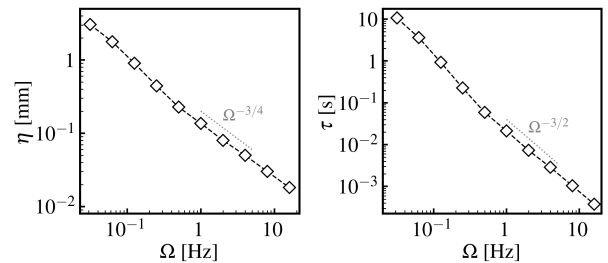


FIG. 5. Kolmogorov length scale and time scale as a function of rotation rate of the agitator blades.

IV. RESULTS

The analysis of turbulence modulation by surfactant addition begins with the turbulent intensity at the bulk region, which is described here by the spatially averaged RMS of the velocity fluctuations. Since we conduct planar measurements, this magnitude is calculated

from the two measured components, by defining $q'_{2D} = \sqrt{\langle u'^2 \rangle + \langle v'^2 \rangle} = \sqrt{u_{\text{rms}}^2 + v_{\text{rms}}^2}$, which is hereinafter referred to as q' for simplicity. The results are then spatially averaged across the measurement region, since these statistics are approximately homogeneous in this area as previously discussed. Note that although the value of q' does not contain the out-of-plane component of the turbulent kinetic energy, its magnitude can be expected to be similar to the currently measured horizontal component of the velocity fluctuations. With this in mind, the results of q' are interpreted as an indication of the trends of turbulent kinetic energy in the bulk region. Aiming to observe the modulation of such a quantity, figure 6 then presents the ratio of the spatially averaged velocity fluctuation RMS between the surfactant and Newtonian cases.

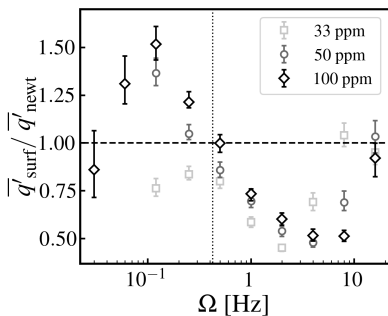


FIG. 6. Ratio of spatially averaged velocity fluctuation RMS between the surfactant and Newtonian cases. Vertical dashed line represents the rotation rate for which $Wi \approx 1$ considering a relaxation time of $\tau_s = 0.1$ s. Error bars represent 95% confidence interval.

According to the classical picture of turbulence modulation via the addition of surfactants and polymers, eddies with a turnover time shorter than the relaxation time of the viscoelastic solutions should be attenuated, and the turbulence in the regime where $Wi < 1$ should remain approximately unchanged. The results shown in figure 6, however, display a more complex picture. It can be seen that although turbulent energy is indeed attenuated once the Weissenberg number crosses the frontier of unity, for all cases, the turbulence also appears to be intensified for the 50 ppm and 100 ppm solutions at low rotation rates. Furthermore, once the rotation rate is increased past a certain threshold, the attenuation is lost and the solutions appear to return to the Newtonian behavior.

To help us understand the reason behind these phenomena, we can start by looking at the effect of surfactant addition on mean flow intensity, which can be defined as the average flow velocity at a given region around the stagnation point. Considering the symmetric nature of the experimental apparatus, as well as the forcing with respect to the right and left regions of the measurement domain, we can enforce this symmetry on the mean flow to improve convergence of the results. This is done by defining a mirror coordinate system as $\tilde{\mathbf{x}} \equiv (2x_c - x, y)$,

with $x = x_c$ corresponding to the symmetry axis, or stagnation point. This then allows us to write a symmetry-enforced mean flow field component, defined as:

$$U^\dagger(\mathbf{x}) = \frac{1}{2} [U(\mathbf{x}) - U(\tilde{\mathbf{x}})]. \quad (3)$$

We can then compute the spatial average of this quantity to then obtain a representative intensity of the mean flow. Since the field is antisymmetric however, this spatial average is computed only on the left-hand side of the point (such that $x < x_c$), resulting in the quantity $\bar{U}_L^\dagger$. Note that this approach is chosen (in opposition to taking the spatial average of the velocity magnitude) because it allows us to preserve the flow direction, and thus take into account flow reversal as well should it happen. Moreover, since the defined quantity has the significance of “spatially averaged mean flow” in the bulk region, it will hereinafter simply be referred to as $\bar{U}$ for visual clarity. With this in mind, figure 7 presents the ratio of mean flow intensity between the surfactant and Newtonian cases.

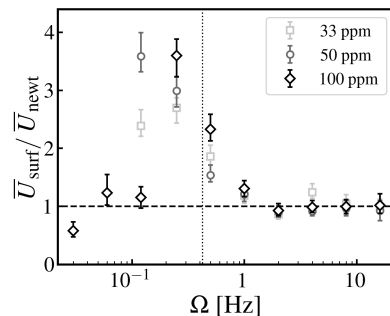


FIG. 7. Ratio of spatially averaged horizontal velocity magnitude between the surfactant and Newtonian cases. Vertical dashed line represents $Wi \approx 1$. Error bars represent 95% confidence interval.

From the plot of figure 7, it can be seen that for rotation rates of approximately 1 Hz or above, the addition of surfactant has almost no effect on the mean flow, irrespective of the concentration. It can be stated that such would be the expected behavior for turbulence which is inertially driven, since the role of surfactant additives is traditionally understood to be the attenuation of the small scales of the flow. Since the driving rotors act on the large scales, which even at the slowest rotation rate are larger than the Kolmogorov length scale by a factor of at least one decade (ref. fig. 5), it can be expected that such large scales should remain mostly unaffected. Nevertheless, the results from figure 7 indicate that reducing the rotation rate leads to an increase of up to 4 times in the mean flow intensity for the regime where $\Omega \lesssim 1$ Hz. Additionally, this behavior appears to be independent of the frontier of $Wi \approx 1$, with the peak in mean flow intensification occurring well below the value of unity of the Weissenberg number, after which the amplification

effect appears to cease once the rotation rate is further decreased.

To investigate the cause behind this increase in the mean flow for the lower rotation rates, figure 8 presents the value of the same spatially averaged horizontal component of the mean velocity, but as a function of the phase of the blade (ϕ). Similarly to the previously defined quantities, we introduce $U_\phi^\dagger(\mathbf{x}, \phi) = \frac{1}{2} [U_\phi(\mathbf{x}, \phi) - U_\phi(\tilde{\mathbf{x}}, \phi)]$ corresponding to the symmetry-enforced version of the phase-locked component of the mean flow U_ϕ . Again, for visual clarity, this quantity is hereinafter simply referred to as $\overline{U}_\phi$ after the spatial averaging over the left side of the observed domain is taken.

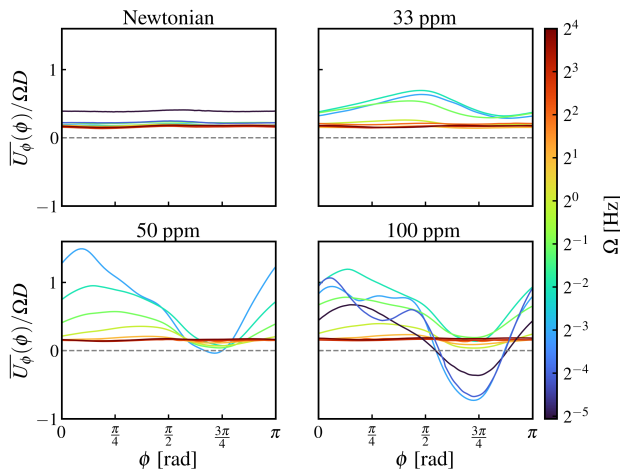


FIG. 8. Spatially averaged horizontal mean flow component as a function of blade phase for each of the tested cases. Positive value refers to inward motion (towards the stagnation point), while a negative value indicates flow reversal.

From these plots, it is possible to see that the overall dependence of the mean flow on the phase of the blade gets amplified as either the concentration of the solution is increased or the rotation rate is decreased, possibly leading to a reversal of flow direction as well. Looking in more detail, from the plots of the 50 and 100 ppm solutions, it can be seen that as the rotation rate is decreased from a high value, the oscillatory behavior is asymmetric, in the sense that we see a strong increase in the mean flow intensity for most of the cycle, and only a small decrease during a very narrow range. This behavior reaches an extreme around the rotation rate of 2^{-2} or 2^{-3} Hz (depending on solution concentration) where the intensification of the mean flow reaches a peak. To aid the analysis, figure 9 presents the normalized amplitude of the curves for each of the cases.

The amplitude curve for each of the solutions indicates that while the Newtonian flow displays an approximately constant normalized amplitude, the surfactant solutions show the presence of a peak, or frequency, at which the oscillations are most pronounced. While data for the slowest rotation rates of the 50 ppm solution is not available, analysis of the curve shapes from figure 8 indicates

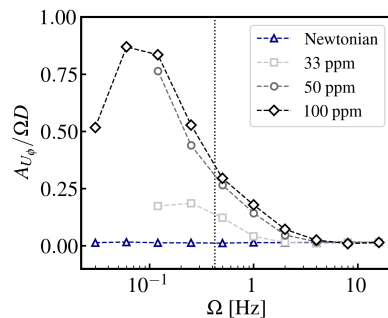


FIG. 9. Normalized amplitude of the spatially averaged, phase-locked horizontal component of the mean flow as a function of rotation rate.

that the oscillations grow in a similar manner between the cases, and that the oscillation peak of the 50 ppm solution should happen at a lower frequency than the 100 ppm counterpart.

Returning to the plots of figure 8 once again, it can be seen that for the 100 ppm solution, where the two slowest rotation rates were tested, as the rotation rate is further reduced the oscillations then tend to become more symmetric, with the appearance of a pronounced region of flow reversal. Even though the rotation of the blades is always in the sense of pushing flow towards the central region (as seen from the point of view of the camera), an opposite movement is observed, with the flow going from the center towards the lateral edges of the domain. On the opposite end of the spectrum, it can be seen from both plots that past a certain rotation rate (approximately 4 Hz in the current experiment), the oscillations cease, and the three solutions recover the Newtonian behavior, similarly to what is seen in the plot of figure 7.

The behavior observed in figures 8 and 9 is indicative of a resonance phenomenon between the flow forcing and large structures which have a relaxation time much slower than the typical relaxation time of the micelles, which, as previously described, is of the order of 0.1 s. Considering that the current resonant peaks appear at a rotation rate of approximately 0.1 Hz (and therefore a forcing frequency of 0.2 Hz due to the geometry of the blades), it can be expected that the structures at play in this case also have a relaxation time of the order of 5 s. Moreover, for them to interact with the blades and produce the observed scale of velocity oscillation, the spatial scale of these structures can also be expected to be of the order of $\mathcal{O}(1\text{cm})$, the same as the spacing between the blades.

In order to analyze the movement and behavior of such flow structures in more detail, we can rely on the flow fields measured in the horizontal plane and in the vicinity of the blades. Considering the limited viewing area enforced by the proximity of the blades, and the maximum mirror size that can be fitted between them, the schematic shown in figure 2 is slightly modified such that

the line of sight of the camera points to the region between two blades, in the orientation at which the motion of “pushing” fluid towards the stagnation point is targeted. In preparation for the analysis of the results, figure 10 shows the relative position of the measured area, displayed as the black square with the velocity vectors. The cross-sections of the blades are drawn as gray rectangles with the rotation direction specified, and the relative positions of the laser sheet and camera viewing direction for the cases of bulk imaging (used for the extraction of the results discussed up to this point) are also shown for reference. In each of the blades, the points defined as leading and trailing edges are also specified as they will be referred to in the analysis of the results. Note that for clarity reasons only a fraction (15% to 20%) of the tracked particles is shown, and the figures are not representative of the actual spatial resolution of the measurements.

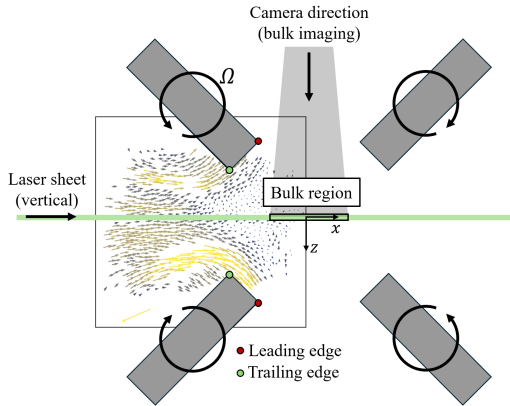


FIG. 10. Schematic of the viewing area in the horizontal plane with respect to the position of the blades, as well as laser sheet and camera directions during bulk imaging.

With this schematic in mind, figure 11 displays the evolution of the flow field over the course of half a rotation period divided into four snapshots, for the Newtonian and 50 ppm cases at the rotation rate of 0.125 Hz, divided into the top and bottom rows respectively. In both cases, the evolution is displayed from left to right, with the blades initially in the position of largest constriction of the flow. From there, each consecutive snapshot is delayed by a rotation of 45° . For the Newtonian flow case, in the top row, no particularly noteworthy flow structures are visible, with the highest flow velocity seen in the vicinity of the blades (particularly behind the trailing edge), and a general attenuation of the velocity magnitude towards the bulk region. For the surfactant case in the bottom row, however, more complex flow patterns are observed.

Analyzing the flow structures of the surfactant case, in the bottom row of figure 11, we start at the first snapshot where the blade edges are at their closest approach. In this orientation the motion of the blades is accompanied by broader (when compared to the Newtonian flow)

higher velocity regions which extend from the leading edge towards the trailing edge, behind the movement of the blades. In the second snapshot, as the two blades begin to separate, these two faster moving regions transform into a more developed shear-layer structure, with an increase in the flow velocity of the layers in direct proximity to the blade and initial formation of secondary high velocity structures in the space between the blades.

At the third snapshot, the layered structures continue to develop, and the velocity magnitude in the region between the blades also reaches its maximum value. Note that the velocity of these streaks, or jet-like structures, surpasses the tip velocity of the blades by over 4 times, and is strongest at the position of largest separation of the blades. It can be hypothesized that as the leading and trailing edges of the blades drag large micellar networks, these are initially compressed as the blades reach close proximity, and then relax, leading to a pronounced acceleration of the flow in this condition. As these “jets” then impinge into the bulk, this region itself receives lateral compression which relaxes once the jets disappear, leading to the reduction in mean flow intensity and possibly flow reversal in this region.

In the fourth snapshot of figure 11 we can see that the jet-like structures appear to break down, or more simply lose strength, in the measured region, while an increase of velocity in the top and bottom right portions of the domain can be seen, likely signalling that these structures are now being pulled apart and downwards or upwards for each of the blades. As we return to the first snapshot, the constriction of the flow by the two blades rapidly leads to the attenuation of the high velocity regions between the blades, which is also consistent with the rapid drop in velocity which is seen in figure 8.

The streak-like structures shown in figure 11 show resemblance to the elongated structures observed in drag-reduction studies in channel flow using solutions of similar concentrations [19]. In the aforementioned study, the authors argue that the formation of shear-induced structures (SIS) is a key part of the drag-reducing capability of surfactant, and the spatial measurements of flow velocity in the wall-parallel planes of the drag reducing solutions show similar elongated structures in the vicinity of the walls. In our current experiment, there exists a region of mean shear between the blades due to the velocity gradient up to the bulk region, where the mean flow is stagnant. It can be hypothesized that this shear, coupled with the relatively low turbulent intensity of the slower rotation rate cases, leads to favorable conditions for the formation of SIS, which we see as the jet-like structures of figure 11.

For the slower rotation rates (below 1 Hz), the interaction between the blade geometry and the forcing frequency leads to a resonant behavior which drives the acceleration of these structures and an increase in the mean flow. As the shear rate increases further, however, their formation is no longer favored (or the large structures break down) and the mean flow intensity returns to that

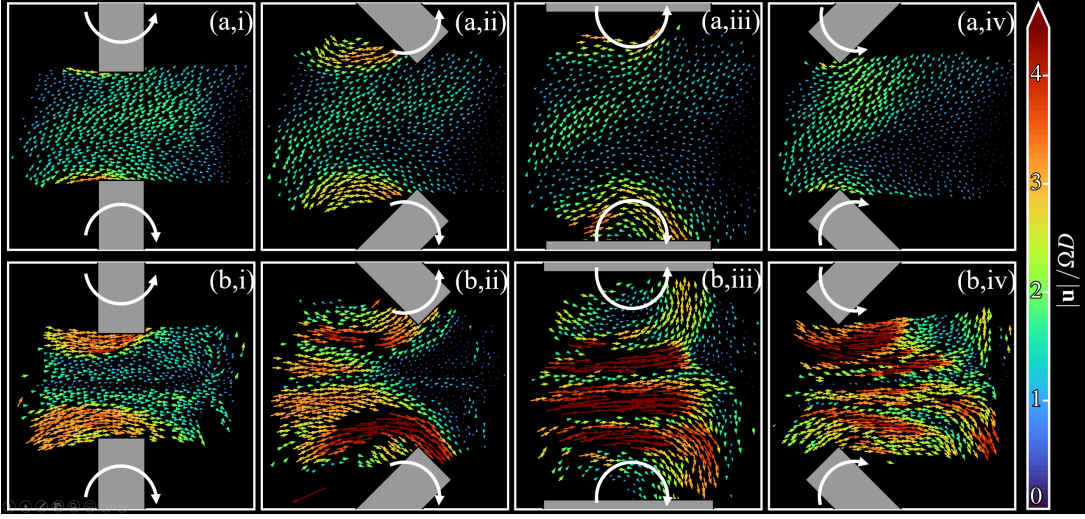


FIG. 11. Top-down visualization of the flow field from Newtonian (a) and 50 ppm (b) cases at rotation rate of 0.125 Hz. For the top and bottom rows, plots (i) through (iv) represent four consecutive snapshots separated by phase difference of $\pi/4$.

of the Newtonian fluid. Moreover, the fact that reducing the rotation rate even further reduces the amplitude of velocity fluctuations is an indication of a mismatch in temporal scales between the rotation rate of the blades and the elasticity of the large structures, which decreases the amplitude of the oscillations and is consistent with the idea of resonance between the large structures and the forcing frequency.

Having established the importance of blade phase on the state of the mean flow in the current experiment, we can now turn our analysis to the evolution of the velocity fluctuations. For this, we apply the triple decomposition technique, which essentially consists of decomposing the flow field into its mean, periodic, and chaotic components [20], in order to remove the contribution of the periodic forcing which would otherwise be included as part of the velocity fluctuations. Essentially, we subtract the phase-dependent mean flow, as shown in figure 8, from the instantaneous flow field in order to obtain the chaotic component of the velocity fluctuations, with magnitude simply defined as (4) to keep consistency with previously defined quantities. The ratio of this spatially averaged, phase-corrected RMS between the surfactant and Newtonian cases is then presented in figure 12.

$$q'_\phi = \sqrt{\left\langle (u(\mathbf{x}, t) - U_\phi(\mathbf{x}, t))^2 \right\rangle + \left\langle (v(\mathbf{x}, t) - V_\phi(\mathbf{x}, t))^2 \right\rangle} \quad (4)$$

When contrasted with the results obtained without the triple decomposition (cf. fig. 6), the results of figure 12 show a less pronounced increase in the turbulent energy for low rotation rates, and the picture becomes closer to what is traditionally expected of turbulence modulation by surfactant addition. Most notably, we see attenuation of the turbulent energy once the Weissenberg number crosses the frontier of unity (approximately), with an at-

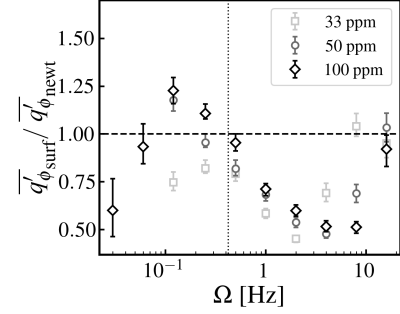


FIG. 12. Ratio of spatially averaged velocity fluctuation RMS (phase-locked) between the surfactant and Newtonian cases. Error bars represent 95% confidence interval.

tenuation which is proportional to the rotation rate in the vicinity of this frontier. In this region, the turnover time of the smallest eddies begins to surpass the relaxation time of the string-like micellar structures, which are then extended and begin to extract energy from these small scales in the process. As the Reynolds number is increased, more turbulent scales begin to interact with the micelles and the attenuation becomes more pronounced. It can also be said that since the attenuation of the turbulent energy does indeed begin to happen around the point where the Weissenberg number was previously estimated to be around unity, this is a good indication that the relaxation time of the solutions is indeed of the order of 0.1 seconds. The presence of other effects, however, such as the resonance with large-scale structures, complicates any comparison of the relaxation time between the three solutions.

Past the attenuation region, as the Reynolds number is further increased, we see another change in the trend of turbulent intensity: the attenuation effect is dimin-

ished, and the turbulent intensity progressively returns to the value of the Newtonian fluid once again. This behavior is analogous to what has been observed in studies of drag reduction in channel flow [19, 21], where a critical Reynolds number exists past which the drag reduction effectiveness sharply declines and the flow returns to the Newtonian state. In such studies, the authors argue that while the presence of some shear is essential for the formation of shear-induced structures which play a key role in the drag reduction, excessive shear leads to the breakup of such structures and inhibits their formation, which undermines the drag reduction effectiveness.

While the current experimental arrangement differs from the case of a channel flow, analogous conclusions can be drawn, and moreover, we can propose that this process happens in two main steps, linked to the distinct regimes of turbulence intensification and attenuation that have been reported. Initially, for very slow rotation rates, the mean shear favors the formation of very large SIS, which then interact with the forcing frequency, leading to the previously discussed resonance phenomenon. As the rotation rate is increased, these structures break down, or their formation is simply not favored, but the individual string-like micellar structures remain intact. In the second regime, the turnover time of the smallest eddies surpasses the relaxation time of these small micellar structures, and the attenuation of turbulence begins. However, as the rotation rate is further increased, and we approach the threshold of $Wi = \mathcal{O}(10^2)$ in the bulk, the shear rate becomes large enough that these smaller micellar structures break down. The resulting smaller micelles should then have much faster relaxation times, and the turbulence attenuation mostly ceases.

Here it is also important to note that we believe the geometry of the blades themselves might play an important role in the rotation rate at which the breakdown happens in both regimes. The rectangular cross-section of the blades, while useful in the sense that it allows us to isolate the inertial component of the flow forcing, also generates regions of very high shear rate in the vicinity of the leading and trailing edges. This in turn means that the Weissenberg number in this region is considerably higher than that of the bulk, and the breakdown of the micellar structures might be concentrated in this region, especially in the case of the second regime at very high rotation rates.

Another possible hypothesis for the behavior observed in the loss of turbulence attenuation is that before the breakdown process begins, the string-like micellar structures are saturated (at maximum extension) and are therefore not able to absorb more energy, even though the energy input rate is being increased with the increase in rotation rate. This hypothesis would then be consistent with the observation that the rotation rate at which the turbulent energy ratio becomes positive again is proportional to the concentration of the solutions, with this process happening first for the 33 ppm solution and being most delayed for the 100 ppm solution. Nevertheless, the

unclear rheological characteristics of the solutions make it difficult to precisely pin down the mechanisms at play here.

A final observation worth noting is that the breakdown of the large micellar structures being aligned with a Weissenberg number close to unity is a coincidence brought forth by the characteristics of the current experiment, and that these phenomena do not necessarily need to be aligned. It is still unclear whether the resonant frequency of these very large scale structures is determined mostly by the geometrical constraints of the experimental device, or simply by the solution itself (most likely), and also how the ratio between the two observed characteristic times changes depending on concentration and solution chemistry.

V. CONCLUSIONS

In this work, we conducted an experimental investigation of the turbulence modulation caused by the addition of a dilute cationic surfactant (CTAC/NaSal) to a Taylor–Green-like forced flow, using blade-shaped rotors to isolate inertial forcing over a wide range of Reynolds numbers and Weissenberg numbers. By combining measurements of the mean flow, phase-resolved analysis of the velocity field, and triple decomposition of the turbulent fluctuations, we were able to show that the turbulence modulation by surfactant addition departs considerably from the classic picture of monotonic attenuation of turbulent energy past a single critical Weissenberg number.

Instead, our results indicate that the response of the flow to the surfactant additive can be divided into three distinct and possibly overlapping regimes as a function of the rotation rate. The first regime, which occurs at the lowest rotation rates ($\Omega \lesssim 1$ Hz), is perhaps the most distant from the classical picture, in the sense that the surfactant does not primarily attenuate the turbulence but rather intensifies the raw velocity fluctuations, together with an intensification of the mean flow. We were able to show that this intensification is tied to a resonance-like interaction between the periodic forcing of the blades and large, elastic micellar structures with a much longer relaxation time (of the order of seconds) than that inferred for the small-scale turbulent fluctuations. This resonance also leads to a strong dependence of mean flow intensity on the agitator phase, and possibly even reversal of flow direction. A direct visualization of the flow field in the vicinity of the blades revealed the mechanism behind this behavior: the periodic compression and relaxation of these large-scale structures by the movement of the blades generates streak- or jet-like structures which accelerate well beyond the tip speed of the agitators, and directly drive the observed intensification of the mean flow in the bulk region.

Once the periodic nature of the velocity fluctuations is properly accounted for via triple decomposition, the re-

sulting phase-corrected velocity fluctuations reveal a picture which is much closer to the traditional one. In the second regime, after the Weissenberg number crosses the approximate value of unity, turbulent energy decays as the turnover time of the smallest eddies becomes shorter than the relaxation time of the string-like micelles, which are then stretched and extract energy from the turbulent cascade. As the Reynolds number increases, so does the attenuation of turbulent energy, until each of the solutions reaches a threshold beyond which the attenuation ceases and the velocity fluctuations recover the behavior of Newtonian flow, characterizing the third regime. In this region, we attribute the loss of turbulence attenuation to the shear-induced breakdown of the micellar network, a behavior analogous to the critical-Reynolds-number effects which have been reported in channel-flow drag reduction studies.

Taken together, our results suggest that the modulation of turbulence in bulk, inertially-driven flow with dilute surfactant solutions cannot be reduced to a single attenuation mechanism governed by a single relaxation time. Rather, at least two distinct groups of viscoelastic structures appear to be relevant, consisting of the large-scale structures which can resonate with the periodic forcing, and the small-scale, string-like micelles which are responsible for the attenuation of small-scale turbulent eddies. We propose a qualitative two-step scenario which connects the effects of these two groups and leads to the resulting modulation of turbulence, but we

also stress that many aspects of this picture remain speculative. In particular, whether the breakdown of the large-scale structures coincides with $Wi \approx 1$ as a consequence of the specific geometry and forcing frequency of the current apparatus, or reflects a more general property of the surfactant solution, cannot be settled with the present data. Likewise, the possibility that the loss of attenuation at high rotation rates is caused by saturation of the micellar extension, rather than by outright structural breakdown, remains an open hypothesis, motivated only indirectly by the observed dependence of this transition on solution concentration.

Future work should aim to disentangle these possibilities and some of the ambiguities in the current observations. For instance, independently varying the geometrical length scale of the experimental apparatus, as well as the geometry of the blades, in a more systematic manner should allow us to pin down whether these parameters modify the relaxation time and breakdown point of the large-scale structures in the first regime. Likewise, changing the chemistry of the solutions should allow us to verify whether there is a direct link between the relaxation time of the small structures and the larger networks, or whether these regimes represent truly independent behaviors. Moreover, considering viscously-driven turbulence should help us clarify whether the mechanisms observed here are specific to inertially-driven turbulence, or are a more general characteristic of surfactant solutions subject to periodic straining.

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